Ismail Kuru, Ph.D.

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Employment History

2026(Feb) –	**Postdoctoral Researcher.** Aarhus University, Department of Computer Science, Center for Basic Research in Program Verification. Working with Prof. Lars Birkedal and Prof. Jean Pichon-Pharabod.
2026(Oct) –	**Postdoctoral Researcher.** Center for R&D on Secure Computer Systems (CRADSEC), Research Organization of Information and Systems (ROIS), Tokyo, Japan. Working with Prof. Taro Sekiyama.
2026(Apr) – 2026(Sep)	**Visiting Researcher.** National Institute of Informatics, Tokyo, Japan. Hosted by Prof. Taro Sekiyama; co-advised a summer research intern.
2025(July) – 2026(Feb)	**Postdoctoral Researcher.** Drexel University, College of Computing & Informatics.
2016(Oct) – 2025(June)	**Doctoral Research Fellow.** Drexel University, College of Computing & Informatics.
2019(Apr) – 2019(June)	**Research Intern.** BedRock Systems.
2016(Jan) – 2016(Sep)	**Senior Software Engineer.** Crytek GmbH.
2014(Sep) – 2015(Feb)	**Research Assistant.** Koc University. Advisor: Dr. Serdar Tasiran.
2014(Apr) – 2014(Aug)	**Research Intern.** Microsoft Research. Advisors: Dr. Matthew J. Parkinson, Ben Hall and Serdar Tasiran.
2013(Sep) – 2014(Mar)	**Visiting Research Student.** University of Washington. Advisor: Dr. Dan Grossman.
2012(Sep) – 2013(Sep)	**Research Assistant.** Koc University.
2011(Mar) – 2011(Sep)	**Software Developer.** GNU Compiler Collection, Google Summer of Code 2011. Advisor: Dr. Albert Cohen.

Education

2018 – 2025 **Ph.D., Drexel University, U.S.A** in Software Reliability.
Thesis: *Modal Abstractions for Operating System Kernels*
Advisor: *Dr. Colin S. Gordon*
Funding: NSF Career Project (*link*)

2016 – 2018 **M.Sc. Computer Science, Drexel University, U.S.A.**
Internships: BedRock Systems
Coursework Completed

2015 **M.Sc. Computer Science, Koc University, Turkey** in Software Verification.
Thesis title: *Static Methods for Checking Correctness of Programs on Relaxed Memory Systems.*
Advisor: *Dr. Serdar Tasiran*
Scholarship: 1 of 18 Microsoft Research Europe PhD Scholarships (*link*)
Internships: University of Washington Visit and MSR Cambridge
Coursework Completed in Technical University of Munich, Germany.

Teaching and Mentoring

Drexel
- **Teaching Assistant**, CS 360 Programming Languages; led discussion sections.
- **Teaching Assistant**, SE 311 Software Engineering.

Koc
- **Teaching Assistant**, Software Engineering, Koc University.

2026
- **Research Advisor**, Keisuke Nishimura (research intern, National Institute of Informatics, summer 2026), program logic for inline assembly in Iris; co-advised with Taro Sekiyama through weekly meetings.

2021 –
- **Long-Term Mentor**, SIGPLAN-M (ACM SIGPLAN mentoring program), mentoring graduate and senior undergraduate students in programming languages research.

Skills

Languages	Strong reading, writing and speaking in English and Turkish. Intermediate German.
SMT Based Verification	VCC, Boogie, QED.
Model Checking	NuSMV.
Type Theory Based Verification	Rocq (formerly Coq), Iris.

Miscellaneous Experience

Awards and Achievements

2024
- **Scholarship**. School on Univalent Mathematics, University of Minnesota Twin Cities, 2024.

2018
- **Scholarship**. DeepSpec Summer School Scholarship Princeton U.S.A, 2018.

2017
- **Scholarship**. PLMW Scholarship for ICFP'17 Oxford U.K, 2017.

2014
- **Travel Grant**. Microsoft Research Visitor Grants(Multiple) Hosted by Dr. Matthew Parkinson, 2014.

2011–2014
- **Graduate School Scholarship**. Received one of 18 Microsoft Research Europe PhD Scholarship awards, 2011-2014.

2012
- **Travel Grant**. from Koc University for VCLA Winter School Vienna, 2012.

2011
- **Travel Grant**. Inria-Paris Visitor Grants(Multiple) Hosted by Dr. Albert Cohen, 2011.

Summer/Winter Schools

2018
- DeepSpec Summer School Princeton U.S.A, 2018.

2017
- PLMW for ICFP'17 Oxford U.K, 2017.
- OPLSS Summer School Oregon U.S.A, 2017.

2012
- ACACES Eighth International Summer School Italy, 2012.
- Microsoft Research Ph.D. Summer School Cambridge U.K, 2012.
- Vienna Center for Logic and Algorithms Winter School on Verification Wien Austria, 2012.

2010
- ACACES Sixth International Summer School Barcelona Spain, 2010.

Talks

2025
- *Principles and Practices for Foundational Operating System Kernel Verification*. Research Organization of Information and Systems (ROIS), Tokyo, Japan, November 2025.

2024
- Princeton PL Group, Princeton University, May 2024.

Miscellaneous Experience (continued)

- UPenn PL Club, University of Pennsylvania, May 2024.
- NJPLS 2024, New York University, May 2024.
- Languages, Systems and Data Seminar, UC Santa Cruz, April 2024.

2023
- NJPLS 2023, University of Pennsylvania, May 2023.

Service and Volunteering

2021
- SIGPLAN-M Long Term Mentor (link)

2024
- OOPSLA'24 Subreviewer.

2020
- POPL'20 Artifact Evaluation Committee.

2019
- OOPSLA'19 Artifact Evaluation Subreviewer.

2018
- PLDI'18 Student Volunteer.

2013
- SEFM'13 Subreviewer.
- SAS'13 Subreviewer.

2012
- RV'12 Subreviewer.

Ongoing Projects

2026 –
- **Reasoning principles for user trusted applications (UTAs)** (with Taro Sekiyama, CRADSEC/ROIS). Program-logic reasoning principles for trusted execution environments, modeled on OP-TEE user trusted applications: the GlobalPlatform TEE protocol for memory shared between a client and a UTA, and isolation enforced by physical memory protection, mechanized in Iris over the Sail RISC-V semantics.

2025 –
- **iIris: a layered Iris program logic for x86-64**, with assembly, macro, and intermediate layers composing through an inline-assembly boundary, a verified encoder/decoder and compiler, and a port of the modal abstractions to a verified xv6 kernel on RISC-V. Nearing completion.

Research Publications

Conference Proceedings

1. Kuru, I., & Gordon, C. S. (2025a). Modal Abstractions for Virtualizing Memory Addresses. *Proceedings of the ACM on Programming Languages*, *9*(OOPSLA2), 2338–2366. https://doi.org/10.1145/3763134
2. Kuru, I., & Gordon, C. S. (2019). Safe Deferred Memory Reclamation with Types [Acceptance Rate 32.6% (28/86).]. *European Symposium on Programming (ESOP 2019)*, *11423*, 88–116. https://doi.org/10.1007/978-3-030-17184-1_4
3. Kuru, I., Matar, H. S., Cristal, A., Kestor, G., & Unsal, O. (2013). Parv: Parallelizing runtime detection and prevention of concurrency errors. In S. Qadeer & S. Tasiran (Eds.), *Runtime verification* (pp. 42–47). Springer Berlin Heidelberg.

Thesis

1. Kuru, I. (2025). *Modal abstractions for operating system kernels* [Doctoral dissertation, Drexel University]. Drexel University. https://doi.org/10.17918/00010934

Books and Chapters

1. Cristal, A., Ozkan, B. K., Cohen, E., Kestor, G., Kuru, I., Unsal, O., Tasiran, S., Mutluergil, S. O., & Elmas, T. (2015). *Verification tools for transactional programs* (R. Guerraoui & P. Romano, Eds.). Springer International Publishing. https://doi.org/10.1007/978-3-319-14720-8_14

Workshop

1. Kuru, I., & Gordon, C. S. (2025c). Modal verification patterns for systems software. *Proceedings of the 13th Workshop on Programming Languages and Operating Systems (PLOS 2025), co-located with SOSP 2025*, 25–33. https://doi.org/10.1145/3764860.3768337
2. Kuru, I., Kulahcioglu Ozkan, B., Mutluergil, S. O., Tasiran, S., Elmas, T., & Cohen, E. (2014). Verifying programs under snapshot isolation and similar relaxed consistency models. *9th ACM SIGPLAN Workshop on Transactional Computing (TRANSACT 2014)*.
3. Matar, H. S., Kuru, I., Tasiran, S., & Dementiev, R. (2014). Accelerating precise race detection using commercially-available hardware transactional memory support. *5th Workshop on Determinism and Correctness in Parallel Programming (WoDet 2014)*.

Manuscripts Under Submission

1. Kuru, I., & Gordon, C. S. (2026a). *Deferred reclamation is one discipline: A scheme-agnostic type system for memory safety, mechanized*.
2. Kuru, I., & Gordon, C. S. (2026b). *ModFs: Modal abstractions for snapshotting copy-on-write filesystems*.
3. Kuru, I., & Gordon, C. S. (2026c). *Resource morphisms for protocol indistinguishability*.
4. Liu, Z., Crawford, F. M., Kuru, I., Pichon-Pharabod, J., & Birkedal, L. (2026). *Modular and tractable verification of a realistic ticket lock on mixed-size Arm in separation logic*.

Manuscripts In Preparation

1. Kuru, I., & Gordon, C. S. (2026d). *specNL: Protocols, resources, contexts and specification from natural language*.
2. Nishimura, K., Kuru, I., & Sekiyama, T. (2026). *A program logic for inline assembly*.

Tech Report

1. Kuru, I., & Gordon, C. S. (2025b). Modal verification patterns for systems. https://arxiv.org/abs/2506.01719
2. Kuru, I., & Gordon, C. S. (2024). Modal abstractions for virtualizing memory addresses. https://arxiv.org/abs/2307.14471
3. Kuru, I., & Gordon, C. S. (2018). Safe deferred memory reclamation with types. http://arxiv.org/abs/1811.11853

References

Available on Request